

Blackbird State

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Abstract

Your abstract.

1 Introduction

The blackbird bipedal needs to store its state information in a gym environment. here is the state that is defined

2 State

2.1 Bipedal State

1. Torso Positions (x, y, z)
2. Torso Angles (q_1, q_2, q_3, w) // they are represented as Quaternions
3. Torso Linear Velocity $(\dot{x}, \dot{y}, \dot{z})$
4. Torso Angular Velocity $(\dot{\rho}, \dot{\phi}, \dot{\gamma})$
5. Joint Positions $(\theta_1 \dots \theta_{10})$
6. Joint Velocities $(\dot{\theta}_1 \dots \dot{\theta}_{10})$
7. Contact States (Torso, footL, footR)

This gives us a total of $6(position/orientation)+6(velocities)+20(jointstates)+3(contactsensor) = 35$ dimensions in our state vector

2.2 How to Incorporate this state

We have altogether 35 different states to observe. I am still figuring a reward function for balancing instead of falling, $R(s)$, before figuring out walking.

References

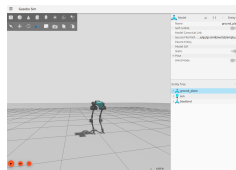


Figure 1: The star of the show